Lesson 4: From Measures to Probability Spaces

AI/ML Mathematics

Contents

1 Motivation and Overview **2**

2 Probability Spaces **2**

3 Random Variables **3**

4 Distributions (Pushforward Measures) **3**

5 Cumulative Distribution Functions **4**

6 Probability Space for a Finite MDP **5**

7 Additional Results **6**

8 Exercises **7**

9 Summary of Key Results **8**

1 Motivation and Overview

Lessons 1–3 developed the general theory of sigma-algebras, measures, and measurable functions. This lesson specializes that framework to *probability*. The key observation is that a probability space is simply a measure space of total mass one, and a random variable is simply a measurable function on that space. This identification lets every theorem from measure theory – Caratheodory extension, the approximation theorem, the Doob–Dynkin lemma – carry over directly.

Goals of this lesson.

1. Define probability spaces as a special case of measure spaces (Kolmogorov axioms).
2. Define random variables, distributions (pushforward measures), and cumulative distribution functions.
3. Prove that the CDF uniquely determines the distribution (via the π-λ theorem).
4. Construct the probability space for a finite Markov decision process, including the natural filtration.

2 Probability Spaces

Definition 2.1 (Probability Space)**.** A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

1. Ω is a non-empty set called the *sample space*,
2. $\mathcal{F}$ is a sigma-algebra on Ω whose elements are called *events*,
3. $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a *probability measure*, i.e., a measure with $\mathbb{P}(\Omega) = 1$.

Explicitly, $\mathbb{P}$ satisfies the *Kolmogorov axioms*:

(K1) $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$.

(K2) $\mathbb{P}(\Omega) = 1$.

(K3) For pairwise disjoint $A_1, A_2, \ldots \in \mathcal{F}$: $\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

Remark 2.2. Every probability measure is a measure in the sense of Lesson 2. Conversely, any finite measure μ with $\mu(\Omega) > 0$ can be normalized: $\mathbb{P}(A) = \mu(A)/\mu(\Omega)$ defines a probability measure. The Kolmogorov axioms are exactly the measure axioms with the additional constraint $\mathbb{P}(\Omega) = 1$.

Theorem 2.3 (Basic Properties of Probability)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A, B \in \mathcal{F}$. Then:*

(a) $\mathbb{P}(\emptyset) = 0$.

(b) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

(c) If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

(d) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

(e) $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$ *(union bound / Boole's inequality).*

Proof. These follow from the corresponding properties of general measures (Lesson 2), but we give self-contained proofs for completeness.

(a) Write $\Omega = \Omega \cup \emptyset \cup \emptyset \cup \cdots$ (pairwise disjoint). By (K3), $1 = \mathbb{P}(\Omega) = \mathbb{P}(\Omega) + \sum_{n=2}^{\infty} \mathbb{P}(\emptyset)$. Since $1 < \infty$, all additional terms must vanish, so $\mathbb{P}(\emptyset) = 0$.

(b) $\Omega = A \sqcup A^c$ (disjoint union), so $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$ by (K3) applied to two sets (extend with $\emptyset$).

(c) $B = A \sqcup (B \setminus A)$, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.

(d) Write $A \cup B = A \sqcup (B \setminus A)$ and $B = (A \cap B) \sqcup (B \setminus A)$. The second gives $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Substituting: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

(e) Define $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n \geq 2$. The B_n are pairwise disjoint with $\bigcup_n B_n = \bigcup_n A_n$ and $B_n \subseteq A_n$. By (K3) and monotonicity: $\mathbb{P}(\bigcup A_n) = \sum_n \mathbb{P}(B_n) \leq \sum_n \mathbb{P}(A_n)$. $\square$

Theorem 2.4 (Continuity of Probability)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.*

(a) ***Continuity from below:*** *If $A_1 \subseteq A_2 \subseteq \cdots$, then $\mathbb{P}(\bigcup_n A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$.*

(b) ***Continuity from above:*** *If $A_1 \supseteq A_2 \supseteq \cdots$, then $\mathbb{P}(\bigcap_n A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$.*

Proof. **(a)** Define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The B_n are pairwise disjoint (if $i < j$ and $\omega \in B_i$ then $\omega \in A_i \subseteq A_{j-1}$, but $\omega \in B_j$ implies $\omega \notin A_{j-1}$, a contradiction). By induction, $\bigcup_{k=1}^N B_k = A_N$, so $\sum_{k=1}^N \mathbb{P}(B_k) = \mathbb{P}(A_N)$ by finite additivity. Taking $N \to \infty$:

$$\mathbb{P}\left(\bigcup_n A_n\right) = \mathbb{P}\left(\bigcup_n B_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{N\to\infty} \mathbb{P}(A_N).$$

(b) If A_n is decreasing, then A_n^c is increasing. By (a), $\mathbb{P}(\bigcup_n A_n^c) = \lim \mathbb{P}(A_n^c) = \lim(1 - \mathbb{P}(A_n))$. By De Morgan, $\bigcup_n A_n^c = (\bigcap_n A_n)^c$, so $1 - \mathbb{P}(\bigcap_n A_n) = 1 - \lim \mathbb{P}(A_n)$, giving the result. □

Connection to RL

Continuity from above is essential for "almost sure" convergence statements. The event "$V_t \to V^\pi$" is $\bigcap_{k=1}^\infty \bigcup_{N=1}^\infty \bigcap_{t=N}^\infty \{|V_t - V^\pi| < 1/k\}$, and computing its probability requires continuity of $\mathbb{P}$ along nested sequences.

3 Random Variables

Definition 3.1 (Random Variable)**.** A *random variable* on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a measurable function $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

More generally, an *S-valued random variable* (or *random element*) is a measurable function $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ for some measurable space $(S, \mathcal{S})$.

Remark 3.2. By the standard criterion (Lesson 3, Corollary 3.2), a function $X : \Omega \to \mathbb{R}$ is a random variable if and only if $\{X \leq x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$. This ensures we can assign probabilities to events like "X is at most x."

Example 3.3 (Indicator Random Variable)**.** For any event $A \in \mathcal{F}$, the indicator function $\mathbf{1}_A : \Omega \to \{0, 1\}$ is a random variable with $\mathbb{P}(\mathbf{1}_A = 1) = \mathbb{P}(A)$ and $\mathbb{P}(\mathbf{1}_A = 0) = 1 - \mathbb{P}(A)$.

Example 3.4 (State as a Random Variable)**.** In a finite MDP with states $\mathcal{S} = \{s_1, \ldots, s_m\}$, the state at time t is a random variable $S_t : \Omega \to \mathcal{S}$. Equipping $\mathcal{S}$ with the discrete sigma-algebra $\mathcal{P}(\mathcal{S})$, measurability is automatic: for every $B \subseteq \mathcal{S}$, $\{S_t \in B\} = S_t^{-1}(B) \in \mathcal{F}$.

4 Distributions (Pushforward Measures)

Definition 4.1 (Distribution / Pushforward Measure)**.** Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (S, \mathcal{S})$ be a random element. The *distribution* (or *law*) of X is the probability measure μ_X on $(S, \mathcal{S})$ defined by

$$\mu_X(B) = \mathbb{P}(X \in B) = \mathbb{P}\big(X^{-1}(B)\big) \quad \text{for all } B \in \mathcal{S}. \tag{1}$$

This is also called the *pushforward* of $\mathbb{P}$ by X, written $\mathbb{P} \circ X^{-1}$ or $X_\# \mathbb{P}$.

Proposition 4.2 (μ_X is a Probability Measure)**.** *The pushforward μ_X defined in (1) is indeed a probability measure on $(S, \mathcal{S})$.*

Proof. We verify the three Kolmogorov axioms.

(K1) Non-negativity: $\mu_X(B) = \mathbb{P}(X^{-1}(B)) \geq 0$ since $\mathbb{P}$ is non-negative.

(K2) Normalization: $\mu_X(S) = \mathbb{P}(X^{-1}(S)) = \mathbb{P}(\Omega) = 1$.

(K3) Countable additivity: Let $B_1, B_2, \ldots$ be pairwise disjoint in $\mathcal{S}$. Then $X^{-1}(B_1), X^{-1}(B_2), \ldots$ are pairwise disjoint in $\mathcal{F}$ (if $\omega \in X^{-1}(B_i) \cap X^{-1}(B_j)$ then $X(\omega) \in B_i \cap B_j$, contradicting disjointness of B_i, B_j). Therefore:

$$\mu_X\left(\bigcup_n B_n\right) = \mathbb{P}\left(X^{-1}\left(\bigcup_n B_n\right)\right) = \mathbb{P}\left(\bigcup_n X^{-1}(B_n)\right) = \sum_n \mathbb{P}(X^{-1}(B_n)) = \sum_n \mu_X(B_n). \quad \square$$

Remark 4.3. Two random variables X and Y (possibly defined on different probability spaces) have the *same distribution* if $\mu_X = \mu_Y$, i.e., $\mathbb{P}(X \in B) = \mathbb{P}(Y \in B)$ for all Borel sets B. This is written $X \overset{d}{=} Y$ or $\mathrm{Law}(X) = \mathrm{Law}(Y)$.

5 Cumulative Distribution Functions

Definition 5.1 (Cumulative Distribution Function)**.** The *cumulative distribution function* (CDF) of a real-valued random variable X is the function $F_X : \mathbb{R} \to [0, 1]$ defined by

$$F_X(x) = \mathbb{P}(X \leq x) = \mu_X\big((-\infty, x]\big), \quad x \in \mathbb{R}. \tag{2}$$

Proposition 5.2 (Properties of the CDF)**.** *Every CDF $F = F_X$ satisfies:*
(a) F is non-decreasing: $x \leq y \implies F(x) \leq F(y)$.
(b) $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$.
(c) F is right-continuous: $\lim_{y\downarrow x} F(y) = F(x)$.

Proof. **(a)** If $x \leq y$, then $\{X \leq x\} \subseteq \{X \leq y\}$, so $F(x) = \mathbb{P}(X \leq x) \leq \mathbb{P}(X \leq y) = F(y)$ by monotonicity of $\mathbb{P}$.

(b) For the left limit, let $A_n = \{X \leq -n\}$. These sets are decreasing ($A_{n+1} \subseteq A_n$) and $\bigcap_{n=1}^{\infty} A_n = \emptyset$ (no real number is $\leq -n$ for all n, by the Archimedean property). By continuity from above (Theorem 2.4(b)):

$$\lim_{n\to\infty} F(-n) = \lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_n A_n\right) = \mathbb{P}(\emptyset) = 0.$$

Since F is non-decreasing, $\lim_{x\to-\infty} F(x) = \lim_{n\to\infty} F(-n) = 0$.

For the right limit, $A_n = \{X \leq n\}$ is increasing with $\bigcup_n A_n = \Omega$. By continuity from below: $\lim F(n) = \mathbb{P}(\Omega) = 1$.

(c) Let $y_n \downarrow x$ (i.e., y_n decreases to x). Then $\{X \leq y_n\}$ is a decreasing sequence with

$$\bigcap_{n=1}^{\infty} \{X \leq y_n\} = \{X \leq x\}.$$

To see the equality: if $X(\omega) \leq x$, then $X(\omega) \leq y_n$ for all n (since $y_n \geq x$), so ω is in the intersection. Conversely, if ω is in the intersection, then $X(\omega) \leq y_n$ for all n, and taking the limit gives $X(\omega) \leq x$. By continuity from above: $\lim F(y_n) = \mathbb{P}(\bigcap_n \{X \leq y_n\}) = \mathbb{P}(X \leq x) = F(x)$. $\square$

CDF Determines Distribution

Proposition 5.3 (CDF Determines Distribution)**.** *The CDF F_X uniquely determines the distribution μ_X. That is, if X and Y are random variables with $F_X(x) = F_Y(x)$ for all x, then $\mu_X = \mu_Y$.*

Proof. We apply the uniqueness theorem from Lesson 1 (via the π-λ theorem).

Step 1: Identify a π-system. Let $\mathcal{P} = \{(-\infty, x] : x \in \mathbb{R}\}$. This is a π-system because $(-\infty, x] \cap (-\infty, y] = (-\infty, \min(x, y)] \in \mathcal{P}$.

Step 2: $\sigma(\mathcal{P}) = \mathcal{B}(\mathbb{R})$. The half-lines $(-\infty, x]$ generate the Borel sigma-algebra (see Lesson 1).

Step 3: μ_X and μ_Y agree on $\mathcal{P}$. For every $x \in \mathbb{R}$: $\mu_X((-\infty, x]) = F_X(x) = F_Y(x) = \mu_Y((-\infty, x])$.

Step 4: Apply uniqueness. By the uniqueness corollary of the π-λ theorem (Lesson 1), two probability measures that agree on a π-system agree on the generated sigma-algebra. Since μ_X and μ_Y agree on $\mathcal{P}$, they agree on $\sigma(\mathcal{P}) = \mathcal{B}(\mathbb{R})$. Hence $\mu_X = \mu_Y$. □

Remark 5.4. This result shows that the CDF is a complete description of a random variable's distributional behavior. Two random variables with the same CDF assign the same probability to every Borel set.

Proposition 5.5 (Point Probabilities from the CDF)**.** *For any $x \in \mathbb{R}$,*

$$\mathbb{P}(X = x) = F_X(x) - F_X(x^-) \tag{3}$$

where $F_X(x^-) = \lim_{y \uparrow x} F_X(y)$ is the left limit. In particular, $\mathbb{P}(X = x) > 0$ if and only if F_X has a jump discontinuity at x.

Proof. Let $y_n \uparrow x$ with $y_n < x$. Then $\{X \le y_n\} \uparrow \{X < x\}$ (the increasing union $\bigcup_n \{X \le y_n\}$ equals $\{X < x\}$ because for any ω with $X(\omega) < x$, there exists n with $X(\omega) \le y_n < x$). By continuity from below:

$$F_X(x^-) = \lim_n F_X(y_n) = \lim_n \mathbb{P}(X \le y_n) = \mathbb{P}(X < x).$$

Since $\{X \le x\} = \{X < x\} \sqcup \{X = x\}$:

$$\mathbb{P}(X = x) = \mathbb{P}(X \le x) - \mathbb{P}(X < x) = F_X(x) - F_X(x^-).$$ □

6 Probability Space for a Finite MDP

We now construct the probability space underlying a Markov decision process with finite state and action spaces. This example ties together all the concepts of this lesson and provides the foundation for the formal treatment of reinforcement learning.

Example 6.1 (Probability Space for a Finite MDP)**.** Consider a finite MDP specified by:

- State space $\mathcal{S} = \{s_1, \ldots, s_m\}$.
- Action space $\mathcal{A} = \{a_1, \ldots, a_k\}$.
- Reward set $\mathcal{R} \subset \mathbb{R}$ (finite).
- Transition dynamics: $p(s', r \mid s, a) = \mathbb{P}(S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a)$.
- Policy: $\pi(a \mid s) = \mathbb{P}(A_t = a \mid S_t = s)$.
- Initial distribution: $p_0(s) = \mathbb{P}(S_0 = s)$.

The probability space $(\Omega, \mathcal{F}, \mathbb{P}_\pi)$.

Sample space. A trajectory is an infinite sequence

$$\omega = (s_0, a_0, r_1, s_1, a_1, r_2, s_2, a_2, \ldots).$$

The sample space is the set of all such trajectories:

$$\Omega = (\mathcal{S} \times \mathcal{A} \times \mathcal{R})^\infty.$$

Sigma-algebra. Equip $\mathcal{S}$, $\mathcal{A}$, $\mathcal{R}$ each with the discrete sigma-algebra (the power set). The product sigma-algebra on Ω is

$$\mathcal{F} = \sigma(S_0, A_0, R_1, S_1, A_1, R_2, \ldots),$$

where S_t, A_t, R_{t+1} are the coordinate projection functions.

Probability measure. By the Ionescu-Tulcea extension theorem (the infinite-product version of Caratheodory), there exists a unique probability measure $\mathbb{P}_\pi$ on $(\Omega, \mathcal{F})$ satisfying

$$\mathbb{P}_\pi(S_0 = s_0, A_0 = a_0, R_1 = r_1, S_1 = s_1, \ldots, R_T = r_T, S_T = s_T) = p_0(s_0) \prod_{t=0}^{T-1} \pi(a_t \mid s_t)\, p(s_{t+1}, r_{t+1} \mid s_t, a_t). \tag{4}$$

The consistency of finite-dimensional distributions (each marginal obtained from the next by summing over the added coordinate) ensures that the extension is well-defined.

Definition 6.2 (Natural Filtration of an MDP)**.** The *natural filtration* $(\mathcal{F}_t)_{t\geq 0}$ of the MDP is

$$\mathcal{F}_t = \sigma(S_0, A_0, R_1, S_1, \ldots, A_{t-1}, R_t, S_t). \tag{5}$$

This is a *filtration*: $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots$, and each $\mathcal{F}_t$ represents the information available at time t.

Proposition 6.3 (Filtration is Increasing)**.** *$\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$ for all $t \geq 0$.*

Proof. $\mathcal{F}_{t+1} = \sigma(S_0, A_0, R_1, \ldots, S_t, A_t, R_{t+1}, S_{t+1})$ is generated by a larger collection of random variables than $\mathcal{F}_t$. Formally, every generator of $\mathcal{F}_t$ is also a generator of $\mathcal{F}_{t+1}$, so the generated sigma-algebra satisfies $\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$. □

Connection to RL

The filtration $\mathcal{F}_t$ captures the agent's history at time t. The Markov property says that the conditional distribution of the future given $\mathcal{F}_t$ depends only on S_t. Formally: $\mathbb{P}(S_{t+1} = s' \mid \mathcal{F}_t) = \mathbb{P}(S_{t+1} = s' \mid S_t)$ $\mathbb{P}$-almost surely. This will be stated precisely once conditional expectation is defined (Lesson 6).

7 Additional Results

Proposition 7.1 (Distribution of a Function of a Random Variable)**.** *If X is a random variable and $g : \mathbb{R} \to \mathbb{R}$ is Borel measurable, then $Y = g(X)$ is a random variable with distribution*

$$\mu_Y(B) = \mu_X(g^{-1}(B)) \quad \textit{for all } B \in \mathcal{B}(\mathbb{R}). \tag{6}$$

Proof. $Y = g \circ X$ is measurable as the composition of measurable functions (Lesson 3, Example 3.6). For any Borel set B:

$$\mu_Y(B) = \mathbb{P}(Y \in B) = \mathbb{P}(g(X) \in B) = \mathbb{P}(X \in g^{-1}(B)) = \mu_X(g^{-1}(B)).$$ □

Proposition 7.2 (CDF of a Discrete Random Variable)**.** *If X takes values in a countable set $\{x_1, x_2, \ldots\}$ with $p_k = \mathbb{P}(X = x_k)$, then*

$$F_X(x) = \sum_{k:\, x_k \leq x} p_k. \tag{7}$$

In particular, F_X is a right-continuous step function with jumps of size p_k at each x_k.

Proof. $\{X \leq x\} = \bigsqcup_{k:\, x_k \leq x} \{X = x_k\}$ (disjoint since the x_k are distinct). By countable additivity: $F_X(x) = \sum_{k:\, x_k \leq x} \mathbb{P}(X = x_k) = \sum_{k:\, x_k \leq x} p_k$. □

8 Exercises

Exercise 8.1. Show that $|\mathcal{F}|$ (the number of elements in a finite sigma-algebra on a finite set Ω) is always a power of 2. *Hint:* Show that every finite sigma-algebra is generated by a partition of Ω into *atoms*, where an atom is a minimal non-empty element of $\mathcal{F}$.

Exercise 8.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A_n \in \mathcal{F}$. Prove the *first Borel–Cantelli lemma*: if $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(\limsup_n A_n) = 0$, where $\limsup_n A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$. *Hint:* Use continuity from above and the union bound.

Exercise 8.3. Let $F : \mathbb{R} \to [0, 1]$ be defined by

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ x/2 & \text{if } 0 \le x < 1, \\ 3/4 & \text{if } 1 \le x < 2, \\ 1 & \text{if } x \ge 2. \end{cases}$$

(a) Verify that F satisfies all three CDF properties (non-decreasing, correct limits, right-continuous at every point).
(b) Compute $\mathbb{P}(X = 1)$, $\mathbb{P}(X = 2)$, $\mathbb{P}(0.3 \le X < 0.7)$.
(c) Does X have a density on $[0, 1)$? Explain.

Exercise 8.4. Construct a probability measure on $(\{0, 1\}^{\mathbb{N}}, \mathcal{F})$ corresponding to an infinite sequence of fair coin flips, where $\mathcal{F}$ is the sigma-algebra generated by the cylinder sets. Verify $\mathbb{P}(\Omega) = 1$ and countable additivity on cylinder sets.

Exercise 8.5. Let $U \sim \text{Uniform}[0, 1]$ (i.e., $\mathbb{P}(U \le x) = x$ for $x \in [0, 1]$). Let $Y = -\ln(1 - U)$.
(a) Compute the CDF of Y.
(b) Identify the distribution of Y.
(c) Generalize: for a continuous CDF F with inverse F^{-1}, show that $X = F^{-1}(U)$ has CDF F.

Exercise 8.6. Prove the *Monotone Class Theorem*: if $\mathcal{A}$ is an algebra on Ω, then the smallest monotone class containing $\mathcal{A}$ equals $\sigma(\mathcal{A})$. *Hint:* Adapt the proof of the π-λ theorem from Lesson 1.

Exercise 8.7 (RL Application: MDP Filtration)**.** Consider an MDP with $\mathcal{S} = \{s_1, s_2, s_3\}$ and $\mathcal{A} = \{a_1, a_2\}$.
(a) Write out explicitly all elements of $\sigma(S_0)$.
(b) Write out all elements of $\sigma(S_0, A_0)$ as a sigma-algebra on the appropriate product space.
(c) Verify that $\sigma(S_0) \subseteq \sigma(S_0, A_0)$.
(d) For a value function $V : \mathcal{S} \to \mathbb{R}$, verify that $V(S_0)$ is $\sigma(S_0)$-measurable by identifying the Borel function g in the Doob–Dynkin lemma.

9 Summary of Key Results

Concept	**Definition / Statement**	**Role**
Probability space	$(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$	Foundation of all RL theory
Random variable	Measurable $X : \Omega \to \mathbb{R}$	States, actions, rewards
Distribution	Pushforward $\mu_X(B) = \mathbb{P}(X \in B)$	Characterizes randomness of X
CDF	$F_X(x) = \mathbb{P}(X \leq x)$	Right-continuous, non-decreasing
CDF determines law	$F_X = F_Y \implies \mu_X = \mu_Y$	Via π-λ uniqueness
MDP space	Product space with chain-rule measure	Formal setting for RL
Filtration	$\mathcal{F}_t = \sigma(S_0, \ldots, S_t)$	Information at time t

Next lesson: Probability Spaces, Random Variables, and Expectation – we define expectation via the Lebesgue integral, prove linearity, introduce common discrete distributions, and develop Bayes' theorem and conditional probability.

References